

## TECHNICAL BULLETIN

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### Dixie Chemical Amine 221 (DCA 221) **Polyglycoldiamine** **CASRN 4246-51-9**

#### Typical Properties

| Property                    | Value                     | Test Method |
|-----------------------------|---------------------------|-------------|
| Specific Gravity@25°C, g/cc | 1.000-1.020               | D-102       |
| APHA Color, Pt-Co           | 100 Max.                  | D-103       |
| Appearance/Odor             | Clear Liquid, Slight Odor | D-104       |
| DCA 221, %                  | 98.0 Min                  | D-145       |
| Water, %                    | 0.5 Max                   | D-145       |
| Viscosity @ 25 °C, cps      | 9.5-15.8                  | D-150V-25   |
| Amine Equivalent, MEQ/g     | 8.60-9.16                 | D-145D      |
| Flash Point, PMCC, ASTM D93 | 331°F                     | ASTM D93    |

#### Applications

DCA 221 is used as a curing agent in performance epoxy resins systems.

#### Formulating Advantages with DCA 221

- Balance of properties
  - Imparts high impact resistance to epoxy systems.
  - Has low temperature thermal shock resistance.
  - Provides excellent adhesion, steel-to-steel and aluminum-to-aluminum.
- Product quality.
  - Permanent, built-in flexibility.
  - Light color.
  - Low viscosity.
  - Provides low viscosity and moderate pot life when properly combined with liquid Bisphenol A-based resins

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- Versatility of application.
  - Adhesives
  - Tooling, electrical potting and encapsulation, and troweling compounds.  
Working time in coatings can be further extended by the addition of a ketone solvent (10-15% on resin weight).

### TYPICAL STARTING FORMULATIONS AND PROPERTIES

#### CASTINGS (Limited to small volume unless highly filled) High Impact Resistance

|                                     |            |
|-------------------------------------|------------|
| <b>Bis-A, Type II</b>               | 100 pbw    |
| <b>Polyglycoldiamine</b>            | 29 pbw     |
| <b>Pot Life (100g.-mass)</b>        | 40 min.    |
| <b>Flexural Strength at 25°C</b>    | 12,300 psi |
| <b>Tensile Strength at 25°C</b>     | 8,500 psi  |
| <b>Tensile Elongation</b>           | 3.5%       |
| <b>Izod Impact, ft-lb/in. notch</b> | 1.5        |
| <b>Gardner Impact, 6-mil film</b>   | 130 in-lb  |
| <b>Heat Deflection Temperature</b>  | 55°C       |

#### Highly Flexible Compounds

|                              |         |
|------------------------------|---------|
| <b>Bis-A, Type II</b>        | 88 pbw  |
| <b>Butyl Glycidyl Ether</b>  | 2 pbw   |
| <b>Triphenyl Phosphite</b>   | 10 pbw  |
| <b>Polyglycoldiamine</b>     | 26 pbw  |
| <b>Pot Life (100 g-mass)</b> | 40 min. |

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### FLOORING APPLICATIONS Red Troweling Compound (High Impact, Good Chemical Resistance)

#### Part A

|                              |          |
|------------------------------|----------|
| <b>Bis-A, Type II</b>        | 100 pbw  |
| <b>Glycol ether solvent</b>  | 10 pbw   |
| <b>Titanium Dioxide</b>      | 10 pbw   |
| <b>Red Iron Oxide</b>        | 5 pbw    |
| <b>Silica, 100 mesh</b>      | 500 pbw  |
| <b>Pot Life (100 g-mass)</b> | one hour |

#### Part B

|                          |        |
|--------------------------|--------|
| <b>Polyglycoldiamine</b> | 15 pbw |
| <b>EH-30 or DMP-30</b>   | 5 pbw  |

#### Adhesives

|                                    |        |         |
|------------------------------------|--------|---------|
| <b>Bis-A, Type II</b>              | 100pbw | 100 pbw |
| <b>Polyglycoldiamine</b>           | 29 pbw | 29 pbw  |
| <b>Union Carbide® A-187 Silane</b> | ---    | 1.3 pbw |
| <b>Mineral Filler</b>              | 30     | 30      |
| <b>Thixotrope</b>                  | 5 pbw  | 5 pbw   |

#### Cured 7 days at 25°C (77°F)

|                                           |           |           |
|-------------------------------------------|-----------|-----------|
| <b>Lap Shear Strength, Al-Al</b>          | 2,000 psi | 3,590 psi |
| <b>After 7 days immersion 100°F water</b> | 2,434 psi | 3,520 psi |

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#### Properties of Bis-A, Type II Epoxy/Polyglycoldiamine

| Physical Properties                           | Room Temperature Cure | Postcured |
|-----------------------------------------------|-----------------------|-----------|
| Heat Distortion Temperature, °C               | 66                    | 70        |
| Rockwell hardness (M Scale)                   | 51                    | 57        |
| Izod Impact, ft/lb/inch                       | 1.53                  | 1.49      |
| Flexural Strength at 23°C                     | 12,800                | 12,300    |
| Flexural Modulus, psi x 10 <sup>-6</sup>      | 0.40                  | 0.36      |
| Tensile Strength at 23°C, psi                 | 8,880                 | 8,500     |
| Tensile Modulus, psi x 10 <sup>-6</sup>       | 0.39                  | 0.29      |
| Gardner Impact on 6-mil films, inch-lb passed | 130                   | 130       |

#### Electrical Properties

DCA 221 is highly recommended for a wide range of potting, impregnation, and coating applications for moderate temperature electrical service. It offers the following advantages:

- Low viscosity
- Very low vapor pressure
- High boiling point
- Permanent flexibility
- Excellent thermal shock resistance
- Excellent electrical properties to 60°C (140°F)

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Proper formulation can provide faster cure and improved electrical characteristics at elevated temperatures. DCA 221 has excellent wetting properties with fillers, metal, and glass. As a hardener and reactive diluent, it offers exciting possibilities in the development of tailor-made electrical formulations.

#### Bis-A, Type II Epoxy/Polyglycoldiamine DCA 221

|                                |     |
|--------------------------------|-----|
| <b>Arc Resistance, seconds</b> | 110 |
|--------------------------------|-----|

|                                         |     |
|-----------------------------------------|-----|
| <b>Dielectric Strength, S/T at 23°C</b> | 496 |
|-----------------------------------------|-----|

|                              |     |
|------------------------------|-----|
| <b>Volts/mil S/S at 23°C</b> | 462 |
|------------------------------|-----|

|                    |     |
|--------------------|-----|
| <b>S/T at 60°C</b> | 517 |
|--------------------|-----|

|                    |     |
|--------------------|-----|
| <b>S/S at 60°C</b> | 439 |
|--------------------|-----|

|                                                                   |                  |
|-------------------------------------------------------------------|------------------|
| <b>Volume Resistivity at 23°C<br/>megohm-cm, 50% R.H. at 60°C</b> | $>6 \times 10^7$ |
|-------------------------------------------------------------------|------------------|

|                                                          |                  |
|----------------------------------------------------------|------------------|
| <b>Surface Resistivity at 23°C<br/>megohms, 50% R.H.</b> | $>2 \times 10^7$ |
|----------------------------------------------------------|------------------|

|                                    |      |
|------------------------------------|------|
| <b>Dielectric Constant at 23°C</b> |      |
| 60 cycles                          | 4.99 |
| 103 cycles                         | 4.94 |
| 106 cycles                         | 3.90 |

|                                    |      |
|------------------------------------|------|
| <b>Dielectric Constant at 60°C</b> |      |
| 60 cycles                          | 5.12 |
| 103 cycles                         | 5.08 |
| 106 cycles                         | 4.44 |

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#### Electrical Properties of Bis-A, Type II epoxy Cured Using Various Hardeners

| Property                                       | ASTM<br>Test Method | BIS-A, Type II<br>Epoxy \ Commercial Grade Diethylene<br>Triamine | Bis-A, Type II<br>Epoxy/DCA<br>221 |
|------------------------------------------------|---------------------|-------------------------------------------------------------------|------------------------------------|
| Arc Resistance, Seconds                        | D 495               | 110                                                               | 110                                |
| Dielectric Strength, volts/mil                 | D 149-55T           |                                                                   |                                    |
| S/T at 23°C                                    |                     | 420                                                               | 496                                |
| S/S at 23°C                                    |                     | --                                                                | 462                                |
| S/T at 60°C                                    |                     | --                                                                | 517                                |
| S/S at 60°C                                    |                     | --                                                                | 439                                |
| Volume Resistivity,<br>Megohm-cm, 50% RH       | D 257-58            |                                                                   |                                    |
| at 23°C                                        |                     | $2 \times 10^{16}$                                                | $>6 \times 10^7$                   |
| at 60°C                                        |                     | $6 \times 10^{14}$                                                | $>2.6 \times 10^7$                 |
| Surface Resistivity, Megohms,<br>50% RH @ 23°C | D 257-58            |                                                                   |                                    |
|                                                |                     | $>2 \times 10^7$                                                  | $>2 \times 10^7$                   |
| Dielectric Constant at 23°C                    | D 150-54T           |                                                                   |                                    |
| 60 cycles                                      |                     | 4.4                                                               | 4.99                               |
| 103 cycles                                     |                     | 4.2                                                               | 4.94                               |
| 106 cycles                                     |                     | 4.2                                                               | 3.90                               |
| Dielectric Constant at 60°C                    | D 150-54T           |                                                                   |                                    |
| 60 cycles                                      |                     | 4.9                                                               | 5.12                               |
| 103 cycles                                     |                     | 4.4                                                               | 5.08                               |
| 106 cycles                                     |                     | 4.5                                                               | 4.44                               |
| Dissipation Factor at 23°C                     | D 150-54T           |                                                                   |                                    |
| 60 cycles                                      |                     | 0.03                                                              | 0.0038                             |
| 103 cycles                                     |                     | 0.04                                                              | 0.0011                             |
| 106 cycles                                     |                     | 0.01                                                              | 0.0547                             |
| Dissipation Factor at 60°C                     | D 150-54T           |                                                                   |                                    |
| 60 cycles                                      |                     | 0.016                                                             | 0.0079                             |
| 103 cycles                                     |                     | 0.01                                                              | 0.0050                             |
| 106 cycles                                     |                     | 0.01                                                              | 0.0554                             |

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